❶ Specify the number of dimensions to be *4*.

For the Thompson sampling policy, we use the following code:

```
ts = botorch.generation.MaxPosteriorSampling(model, replacement=False)
next_x = ts(candidate_x, num_samples=batch_size)
```

For the Max-value Entropy Search we use the following code, which uses the helper function `optimize_acqf_cyclic()` to implement cyclic optimization. Note that we are specifying that cyclic optimization should only take at most *5* iterations.

```
policy = botorch.acquisition.max_value_entropy_search.qMaxValueEntropy(
    model, candidate_x
)

next_x, acq_val = botorch.optim.optimize_acqf_cyclic(
    policy,
    bounds=bounds,
    q=batch_size,
    num_restarts=40,
    raw_samples=100,
    cyclic_options={"maxiter": 5} ❶
)
```

❶ Specify the maximum number of iterations in cyclic optimization.

6. Plot the optimization progress of the BayesOpt policies we have run and observe their performance.

   Figure A.17 shows the optimization results obtained by the policies we implement above. We see that most policies are comparable except for Thompson sampling; Max-value Entropy Search has a slight edge.

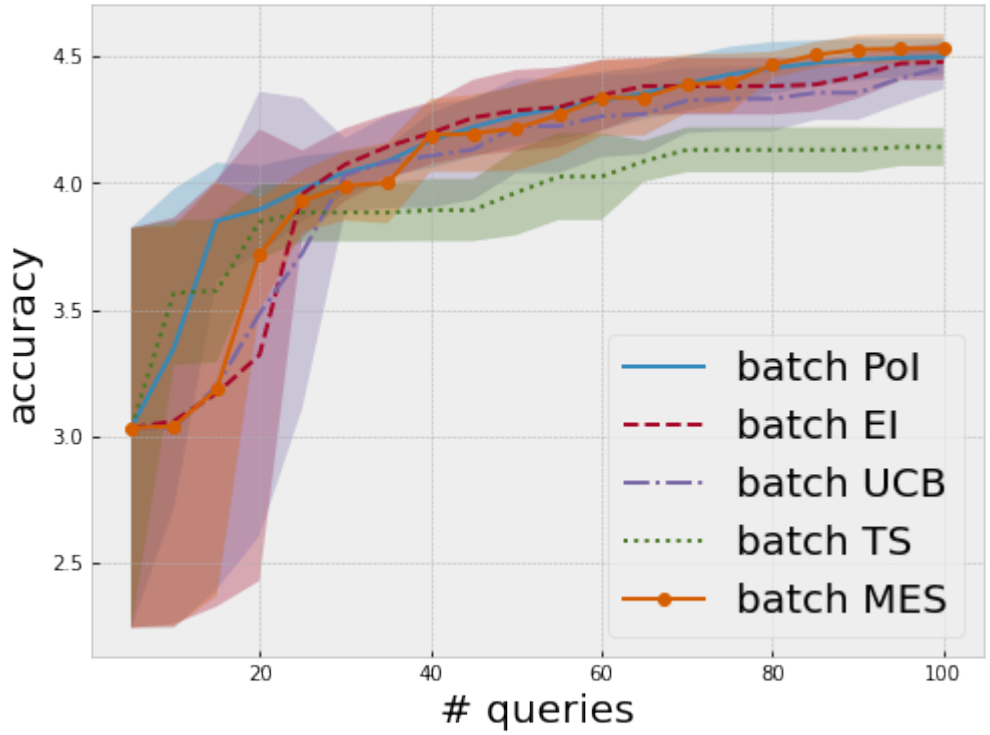


Figure A.17 Progress made by various Bayesian optimization policies in the airplane design optimizatior example. Most policies are comparable except for Thompson sampling; Max-value Entropy Search has a slight edge.

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